

Table 2

Tube roughnesses.

Material	Grit	Roughening method	Average mean roughness across roughness grain R_a [μm]	Average mean roughness across roughness grain R_a [μm]
Copper	G1200	Hand	0.12	0.11
	G600	Hand	0.35	0.35
	G100	Hand	0.74	0.47
	G40	Hand	1.37	0.85
Mild steel	G1200	Angle grinder	0.11	0.10
	G100	Angle grinder	1.91	0.70
Stainless steel	G1200	Angle grinder	0.09	0.10
	G100	Angle grinder	1.32	0.81